Universitat Jaume I

Bachelor's Thesis in Video Game Design and Development

Micorrizas

Design and development of a geolocated mobile video game to promote knowledge and relationship with the natural environment

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ABSTRACT

This document presents the Bachelor's Thesis developed for the Bachelor's Degree in Video Game Design and Development. The project originates from a pedagogical activity carried out with Early Childhood Education students in the course MI1815 – The Natural Environment in Early Childhood Education, consisting of a guided visit to the *Jardín de los Sentidos*, an outdoor space at Jaume I University characterised by its rich biodiversity. The educational value of this activity lies in encouraging students to observe, discuss, and reflect on natural ecosystems through direct interaction with the environment. Building upon this experience, this project explores how educational objectives of the original activity can be transferred to a digital, autonomous and playful format. To address this challenge, *Micorrizas* was designed as an educational video game that promotes environmental awareness and engagement through active exploration and collaborative learning. The resulting game is a cooperative location-based mobile experience developed with Unity, where players interact with real-world locations while progressing through shared challenges and decision-making processes. To support this experience, the game combines geolocation and multiplayer technologies that enable collaborative gameplay in outdoor environments. The game has also been evaluated by the lecturers responsible for the original educational activity, providing feedback on its suitability as a tool for environmental education in outdoor learning contexts.

KEYWORDS

Serious game - educational game - cooperative multiplayer - geolocation - mobile learning - environmental education - Jardín de los Sentidos.

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INTRODUCTION

CONTEXT

Environmental education is increasingly recognised as an essential component of sustainable development, as it promotes the knowledge, attitudes, and competencies required to understand and address environmental challenges. UNESCO highlights the importance of integrating sustainability and climate action into educational practices, promoting learning experiences that encourage critical thinking and environmental responsibility (UNESCO, 2024). In this context, direct contact with nature has been shown to support environmentally responsible attitudes and behaviours, particularly when learners engage in active and meaningful experiences (Rosa et al. 2023).

One example of this approach is the educational activity carried out in the course MI1815 - The Natural Environment in Early Childhood Education at Jaume I University. The activity takes place in the *Jardín de los Sentidos*, located on the Riu Sec campus and recognized as one of the university's three main botanical areas of interest. The garden is a botanical and cultural space of approximately 1,500 m², organised into six thematic platforms: a welcome area and five areas related to sensory perception. It contains more than 260 plant species belonging to 78 different families. Established in 2004 as both an educational and cultural resource, the garden serves as an outdoor laboratory where future teachers can engage directly with the natural environment.

Through guided exploration of the garden, students observe a wide variety of plant species from different regions of the world, develop an appreciation for botanical heritage, and learn to interpret ecological relationships within a real-world context. The sensory-oriented design of the garden also provides opportunities to connect botanical knowledge with pedagogical practices relevant to early childhood education. In addition, the activity encourages students to design and reflect on educational itineraries in natural settings, highlighting the value of nearby green spaces as accessible learning environments that can support environmental education when access to larger natural areas is limited.

MOTIVATION

Although the *Jardín de los Sentidos* activity provides valuable educational experiences, there are currently several limitations to its impact. As it takes place outdoors and within a specific academic timeframe, its implementation is constrained by the duration of the session, environmental conditions, group availability, and the presence of an expert instructor capable of guiding observations, answering questions, and adapting the experience to the learners' needs. These constraints make it difficult to repeat the activity frequently, extend it to additional groups, or maintain it as an ongoing learning process. Consequently, there is a need to explore alternative approaches that preserve the experiential value of the activity while enabling more autonomous, flexible, and sustained participation.

In this context, serious games and mobile technologies offer a promising approach for extending such learning experiences into more accessible, interactive, and engaging educational formats. Serious games are digital games designed with a main purpose that goes beyond entertainment. Although they preserve essential elements of play, such as rules, challenges, interaction and feedback, their design is oriented towards educational, training, social or behavioural objectives (Hands, 2026; Facchino et al., 2025).

In educational contexts, serious games are especially relevant because they can place learners in an active role. Instead of receiving information passively, players interact with content, make decisions, test hypotheses and progressively construct meaning through experience. This approach is closely related to constructivist and experiential learning perspectives, as the game environment can encourage exploration, problem solving and reflection. In addition, serious games can offer safe spaces in which learners are allowed to make mistakes and observe their consequences without the risks associated with real-world practice (Gómez-Cambronero et al., 2019; Hands, 2026; Facchino et al., 2025).

Another key contribution of serious games is their capacity to provide immediate feedback. The system can respond directly to the player's actions, allowing learners to adjust their strategies, correct errors and continue progressing through an iterative process. This feedback can also support assessment, not as an external interruption of the activity, but as part of the game experience itself. However, the educational value of serious games depends on their design and integration within a broader learning context. Poor interface design, excessive cognitive load, technical problems or lack of pedagogical alignment can reduce their effectiveness. Therefore, serious games should be understood as complementary tools rather than replacements for traditional teaching methods (De Gloria et al., 2014; Facchino et al., 2025; Hands, 2026).

The increasing interest in serious games is also linked to technological development. Recent advances in mobile devices, mixed reality, augmented reality, virtual reality, artificial intelligence and learning analytics have expanded the possibilities for creating interactive educational experiences situated in physical or simulated environments. In this context, mobile and location-based games are particularly relevant because they can connect digital interaction with real-world exploration, making the environment itself part of the learning process (Gómez-Cambronero et al., 2019; Hands, 2026).

By combining these approaches, guided outdoor educational activities can be transformed into autonomous and interactive learning experiences in which students engage with challenges, receive feedback, collaborate with peers, and construct environmental knowledge through the exploration of real-world environments.

From this perspective, the present project emerged from a collaboration between the Department of Early Childhood Education and the CEVI research group, to which the supervisor of this Bachelor's Thesis belongs. The aim is to transfer the core learning outcomes of the original activity into a mobile experience that preserves interaction with the real environment while reducing the need for continuous teacher guidance and timetable limitation constraints. To achieve this, *Micorrizas* has been designed with the objective of transforming the exploration of a biodiverse outdoor space into a participatory and motivating experience, guiding students through missions, questions, clues, and contextualized feedback that respond to their actions.

In doing so, the project seeks to support environmental education through a combination of experiential and collaborative learning, serious game design, and mobile technologies. The game guides students through the garden, encourages direct observation of the environment, and requires them to discuss and make decisions as a group. Its narrative, geolocation and cooperative mechanics transform the visit into an interactive experience in which learning emerges from exploration, shared interpretation and immediate feedback. The final group result also provides a basis for reflection on what has been learnt without turning the experience into a conventional test.

This project is also motivated by professional and personal development. From a technical perspective, it represents a challenge because it covers areas I had not previously developed in my university projects, particularly multiplayer interaction and geolocation-based gameplay. The implementation of a serious mobile game that combines map integration, location-based triggers, cooperative synchronisation and mini-games offers the opportunity to expand my skills as a video game developer. At the same time, the project is personally motivating because it connects game design with a real educational context, allowing the final prototype to address a specific need and explore how video games can support learning.

OBJECTIVES

Although the *Jardín de los Sentidos* activity provides valuable educational experiences, there are currently several limitations to its impact. As it takes place outdoors and within a specific academic timeframe, its implementation is constrained by the duration of the session, environmental conditions, group availability, and the presence of an expert instructor capable of guiding observations, answering questions, and adapting the experience to the learners' needs. These constraints make it difficult to repeat the activity frequently, extend it to additional groups, or maintain it as an ongoing learning process. Consequently, there is a need to explore alternative approaches that preserve the experiential value of the activity while enabling more autonomous, flexible, and sustained participation.

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PLANNING AND RESOURCES EVALUATION

This section presents the organisation of the project work. It describes the task breakdown, the estimated workload assigned to each area and the cost analysis derived from the planned development.

PLANNING

To facilitate its development process, the project was divided into a set of tasks, as outlined below. Trello [1] was used to manage the development process, organise pending tasks and track progress throughout the project.

- **Project analysis** (10 hours):
  - **Meetings and research** (6 hours): In order to ensure that the game would adequately meet the educational needs and align with the objectives of the training activity, a number of meetings were held with the teachers involved, both during the initial phase and throughout the course of the project. These sessions made it possible to gather ongoing feedback and guide design and development decisions. The teachers also provided material about the garden, its species and the educational activities carried out there, which required additional research in order to understand the context and adapt the game content appropriately. Furthermore, to address certain technical aspects of the project, contact was maintained with a geolocation specialist, whose advice helped to resolve specific issues relating to this area.
  - **Initial planning** (4 hours): The initial planning phase consisted of defining the scope of the project, identifying the main tasks to be carried out and estimating the time required for each development stage.
- **Game design** (50 hours): The game design phase is an ongoing process, as the project is constantly evolving. A key aspect of game design is ensuring that the focus of the serious game is not lost sight of. This includes ensuring the game is well-balanced, designing cooperative missions, and making sure the game is rewarding for the user, while meeting the pedagogical goals.
- **Art design** (10 hours): This task involved both sourcing and creating the visual and audio assets used in the game. Part of this time was spent searching for references and assets that matched the project's artistic style and narrative, primarily through platforms such as the Unity Asset Store [2] and Itch.io game-assets [3]. It also involved creating or adapting certain necessary resources, such as the character sprites, the images for the Garden of Hearing, and the image representing the mycorrhizae found above each garden. Furthermore, audio resources were explored for the Garden of Hearing quest, including testing recorded sounds and searching specialised sound databases such as Xeno-canto [4], which offers recordings of birdsong and other natural sound material.
- **Development** (120 hours): This was the longest phase of the project and the one with the most iterations. It was divided into several implementation milestones, which are detailed below, including geolocation, multiplayer mode, mission logic and final integration.
  - **Geolocation** (60 hours): Geolocation was the most complex part of the development process. This task required an initial research phase to understand the available location services and their limitations, followed by several rounds of implementation and testing. Its complexity was compounded by the difficulty of validating GPS behaviour outside the development environment, as reliable testing had to be carried out at the actual physical location.
  - **Multiplayer** (20 hours): This task involved researching multiplayer functionality and subsequently implementing the system using the Unity Relay SDK. It included configuring the network service, creating and joining game sessions, as well as ensuring the necessary data synchronisation between players.
  - **Quests** (30 hours): This task focused on designing, implementing and refining the quest system. Each quest had a different narrative and pedagogical purpose, which required specific mechanics, interaction models and feedback systems.
  - **Integration** (10 hours): This task involved bringing together all the previously implemented elements and ensuring they functioned as a single, coherent prototype. Specifically, this involved integrating the geolocation system, the multiplayer functionality, the mission logic and user interaction, as well as testing the application's entire workflow from start to finish in order to iterate and improve it.
- **Testing and iteration** (60 hours): Aquí se ha contabilizado el tiempo de testear cada mecánica de las quest y el tiempo de ir a testear la geolocalización.
- **Documentation** (50 hours): Documentation is an ongoing task, although it cannot be carried out at every step, as the project is constantly evolving. For instance, you might start using an SDK but then encounter difficulties and end up

using a different one, or a task might begin one way and then evolve into different phases.

- **The technical proposal** (5 hours): The technical proposal defined the scope of the project. Its aim was to set out the basic concept of the project, the educational context, the main objectives and the initial planning. This document also helped to clarify important conceptual aspects of the project, such as the distinction between gamification and serious games.
- **GDD** (10 hours): This document defined the core gameplay loop, cooperative mechanics, narrative framework, level structure, interface design, multiplayer model and technical architecture.
- **The Dissertation** (25 hours): The final report is a more comprehensive piece of documentation; for this reason, it was planned to include the theoretical underpinnings of serious games, the rationale behind the project, design decisions, and so on. The previous documents served as the basis for this report, but the content had to be reorganised and expanded from an academic perspective.
- **Project defence** (10 hours): Finally, the presentation was left until the final phase of the project, once all design and implementation decisions had been made. The presentation was a summary of the most relevant aspects of the project.

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Figure 2.1: Project Gantt chart

COSTS

The cost of carrying out this project is broken down below. This does not represent an actual payment, as the project was undertaken as an academic assignment, but it provides an estimate of the professional value of the work carried out

WORKING HOURS

The cost of the work carried out in this project has been estimated according to the number of development hours planned in the Gantt chart. To assign an economic value to these hours, the reference used was the national collective agreement for consultancy and information technology companies, published in the Spanish Official State Gazette (BOE) [5].

The selected professional category was Area 3, Group D, Level 3, since it corresponds to software development and programming tasks and represents an appropriate reference for an entry-level profile in the software industry. According to the updated 2026 salary tables, this category has an annual gross salary of 18,542.34 €. Considering the annual working time of 1,800 hours established in the same agreement, the hourly cost is calculated as follows:

HOURLY COST CALCULATION

$$\frac{18{,}542.34}{1{,}800\ \text{hours}} = 10.30\ \text{/hour}$$

Therefore, the estimated labour cost of the project is:

TOTAL LABOUR COST

$$\text{Total labour cost} = \text{Number of hours} \times 10.30 \quad \text{/hour}$$

If the project required 300 hours of work:

$$300 \text{ hours} \times 10.30 \quad \text{/hour} = 3,090$$

HARDWARE

The development of a similar project would require a computer capable of running the necessary development tools primarily Unity as well as several mobile devices, between two and six devices. We will estimate six devices to be able to test the application under full conditions. The hardware specifications of the system used for this project is described below. OMEN Gaming Laptop 16-apoxxx [6], 32GB of RAM and 954GB of hard disk, NVIDIA GeForce RTX 5070 Laptop GPU. Cost: 1,399€. Xiaomi Redmi Note 13 Pro 5G [7]. Cost: 80€ In order to provide a realistic cost reference to replicate the project, two alternative scenarios can be considered: purchasing a similar device or renting additional test devices. The official Xiaomi reference price for the Redmi Note 13 Pro 5G is approximately 219.99 €. As another option, the company could rent one or two additional Android devices during the validation period. Although the exact Xiaomi Redmi Note 13 Pro 5G was not available in the consulted renting catalogue, similar Android devices were available through Rentik / Phone House [8], such as the Xiaomi Redmi Note 15C from 6.90 €/month, the Motorola G55 5G from 8.90 €/month, or the Xiaomi Redmi Note 14 5G from 16.90 €/month.

The current range is between 6.90 €/month and 16.90 €/month, so an average cost of approximately 10 €/month/device will be estimated. Considering that, during one month, only two devices may be needed for testing, but during another month, up to six devices may be required.Therefore, the estimated cost would be 20 € for the first month plus 60 € for the second month, resulting in a total of approximately 80 €.

SOFTWARE

The software tools and licences required to develop this project were free for educational use, but their commercial value is detailed below for replication in a professional setting:

- **Unity:** Used as the main game engine. While it currently offers a free Personal edition, the pricing structure was recently modified, cancelling the Runtime Fee [9]. For a professional studio, a Unity Pro licence might be required depending on revenue.
- **ArcGIS SDK:** Used for the map and geolocation features. A student use licence [10] is available at no cost for academic purposes, whereas commercial development requires a paid subscription.
- **OpenAI Codex:** Used to assist in the development of certain scripts and technical problem-solving. The pricing model [11] is based on token usage. For this project, the cost was negligible due to the small scale, but it should be considered in larger professional environments.

OTHER EXPENSES

Concept	Cost (€)
Labour (Working hours)	
Hardware	
Software	
Other expenses	
Total	

Table 2.1: General project costs

PLANNING DEVIATIONS

Concept	**Cost (€)**
Labour (Working hours)	
Hardware	
Software	
Other expenses	
Total	

Table 2.2: Planning deviations

CHAPTER 3

ANALYSIS

This section identifies the main requirements that guided the design and development of Micorrizas. The purpose of this analysis is to establish what the system must provide in order to support the intended experience.

The requirements were defined from three main sources: the educational needs of the activity, the game design decisions and the technical limitations of mobile development. The educational context required the game to support biodiversity learning through observation and exploration. The game design required cooperative interaction, narrative progression and feedback. The technical context required geolocation, map integration, multiplayer synchronisation and usability in an outdoor environment.

The requirements are divided into functional and non-functional requirements. Functional requirements describe the actions and behaviours that the system must implement. Non-functional requirements describe the quality conditions that the system must satisfy during use.

FUNCTIONAL REQUIREMENTS

ID	Description
FR01	**Create a game session** The system must allow a player to create a multiplayer session that other users can join.
FR02	**Join a game session** The system must allow players to join an existing session using a code.
FR03	**Register player names** The system must allow players to identify themselves within the group session.
FR04	**Display the game map** The system must show the map of the Jardín de los Sentidos using the selected map integration.
FR05	**Show the player real position** The system must display the approximate position of the player on the map.
FR06	**Detect sensory areas** The system must detect when the player enters the activation area of a sensory garden.
FR07	**Activate quests by location** The system must unlock the corresponding quest when the player reaches the required area.
FR08	**Present narrative content** The system must display narrative dialogue related to the activated area and the current state of the experience.
FR09	**Include one quest per sensory garden** The system must include specific quests for sight, sound, touch, taste and smell.
FR10	**Support cooperative gameplay** The system must require players to collaborate during the quests.
FR11	**Synchronise session state** The system must keep all connected devices updated with the active quest, progress and results.
FR12	**Allow answer submission** The system must allow players to submit answers during the quests.
FR13	**Calculate user performance** The system must calculate a group score after each quest.
FR14	**Calculate the final result** The system must calculate and display the final group result after all required areas have been completed.
FR15	**Provide feedback** The system must provide feedback after each challenge, indicating the result of the group's action.
FR16	**Track completed areas** The system must register which sensory gardens have already been completed.
FR17	**Display final summary** The system must present a final screen with the group members and their final score.
FR18	**Allow replay or restart** The system should allow the group to restart the experience if required.

Table 3.1: Functional requirements

NON-FUNCTIONAL REQUIREMENTS

ID	Description
NFR01	**Mobile platform** The application must run on Android mobile devices.
NFR02	**Outdoor readability** The interface must remain readable in outdoor conditions, including sunlight and variable lighting.
NFR03	**GPS tolerance** Activation areas must be wide enough to compensate for inaccuracies in mobile GPS positioning.
NFR04	**Performance** The application must avoid unnecessary graphical complexity and maintain stable performance on mobile devices.
NFR05	**Loading efficiency** The system should minimise waiting times during map loading, scene transitions and quest activation.
NFR06	**Small group support** The multiplayer system must support small cooperative groups.
NFR07	**Synchronisation consistency** All devices in the same session must receive the same challenge state and result updates.
NFR08	**Simple access** The system should allow players to access a session through a simple code-based flow.
NFR09	**Data minimisation** The application should only handle the information required for the session, such as player names, group progress and score.
NFR10	**Clear interaction** The interface must use simple controls, short instructions and clear feedback.
NFR11	**Maintainability** The content structure should allow new plants, sounds, questions or quests to be added in future versions.
NFR12	**Accessibility support** Audio-based information should include visual or textual support when necessary.
NFR13	**Robustness** The application should remain usable when minor GPS variations or temporary connection delays occur.
NFR14	**Battery awareness** The application should avoid excessive use of continuous processes that could increase battery consumption.

Table 3.2: Non-functional requirements

DEVELOPMENT

GAME CONCEPT

Micorrizas is based on the educational activity carried out in the Jardín de los sentidos as part of the subject related to the natural environment in Early Childhood Education. In this activity, students visit the garden and work directly with the natural space through observation, identification and interpretation tasks. The aim is for students to understand the educational potential of nearby natural environments and to learn about the biodiversity of the garden through direct interaction with plants, sensory areas and ecological elements.

Since the original activity takes place outdoors and requires students to move through the different areas of the garden, Micorrizas was designed as a mobile and geolocated experience. The mobile device makes it possible to guide the players through the physical space, detect their position and activate missions when the group reaches specific areas. In this way, the game preserves the relationship with the real environment while adding a digital layer of interaction, narrative progression and feedback.

The content of the game is organised through missions linked to the senses represented in the garden. Each quest focuses on a specific type of perception or knowledge: visual recognition of plants, identification of sounds, deduction of species from tactile characteristics, the relationship between fruits and species, and classification of plants according to their uses. This structure allows each mission to be connected with a specific area of the garden and with the type of interaction that students would normally carry out during the educational visit.

Cooperation plays a central role in the design. Quests are solved as a group and the score obtained belongs to all members in the same room. Some quests rely on discussion and collective memory, while others distribute information across devices to encourage communication between players. This organisation makes collaboration an active part of how the game functions and reflects the social nature of the original learning activity. Deeproot's narrative and the mycorrhizal network give unity to the experience. Each mission represents the recovery of a part of the garden and contributes to rebuilding the lost connection between the sensory zones. The game's progression is presented as a symbolic restoration of the natural environment, linking players' actions to the observation, interpretation and care of biodiversity.

Some of the decisions I make, which are part of the game concept process. The design of each quest was defined by combining the educational purpose of the original activity with interaction models suitable for mobile devices. For example, the Sight quest is based on a swipe interaction. When a plant card appears on screen, players slide it to the right if they consider that the species has been observed in the Sight area, and to the left if they consider that it does not belong to that area. After each decision, the system provides feedback and updates the group score.

This interaction was chosen because it is simple, fast and well suited to mobile play. The binary gesture reduces interface complexity and allows players to focus on the recognition task. Since only one stimulus is presented at a time, the player's attention remains centred on the plant profile and on the comparison with the real garden. The mechanic also supports short classification rounds, making the activity easy to understand and suitable for outdoor use.

From a pedagogical perspective, the Sight quest reinforces visual discrimination. Players must classify each species by comparing visual evidence, their memory of the physical route and the information shown on the card. The simplicity of the interaction allows the challenge to remain focused on observation and decision-making, while the immediate feedback helps players correct mistakes and progressively refine their understanding of the plants present in the garden.

The Sound mission was designed to connect auditory perception with cooperative interaction. Its structure was inspired by two types of activities: audio-based association exercises, such as Duolingo, and collaborative answer distribution, similar to the dynamic used in Quizlet Live. In each round, one device plays an audio clue related to a sound found in the garden, while the rest of the devices display different possible answers.

Only one of the devices contains the correct answer, so the group must communicate in order to identify it. Players need to listen to the sound, compare the available options and decide together which response should be selected. Once the group answers (correctly or incorrectly), the roles change: another device becomes responsible for playing the sound, while the previous audio device receives answer options. This rotation ensures that the task is distributed among the participants and that all players take part in both listening and decision-making.

From an educational point of view, this mission works on sound recognition, association and group communication. The audio clue connects the digital interaction with the sensory experience of the garden, while the distribution of answers across devices encourages players to share information and coordinate their response. As a result, the mission transforms a simple sound-identification task into a cooperative activity linked to the physical environment.

LORE

Deeproot is the main character in Micorrizas and acts as the game's narrative guide. He is an Ent, a creature directly linked to the flora of the Jardín de los Sentidos, whose role is to protect and manage the area's natural resources. Due to the slowness inherent to his species and the progressive deterioration of the garden, Deeproot seeks the help of a group of adventurers, represented by the players, to restore the lost balance.

The story begins with a conflict caused by the negligence of the human race. For a long time, natural spaces and their biodiversity have been polluted, ignored and displaced, weakening the relationship between people and the environment. In the past, nature flourished throughout the university's grounds, and the Jardín de los Sentidos occupied a central place within that ecosystem. It was the heart of all the flora in the realm and maintained a connection with the rest of the green spaces via an underground network formed by conduits covering the plants' roots.

These conduits were made up of mycorrhizae, the symbiotic relationship between a fungus's mycelium and a plant's roots, and enabled the Jardín de los Sentidos to send nutrients, signals and information to the rest of the flora. Thanks to this network, the plants were able to remain connected, and the garden functioned as a living organism capable of perceiving, remembering and transmitting knowledge through its senses.

At present, this network is fragmented and weakened. The garden's senses have lost part of their memory, and Deeproot can no longer transmit information correctly via the mycorrhizae. Even so, the network can still be restored. To achieve this, players must explore the Jardín de los Sentidos, complete the various missions associated with each sensory zone, and help restore the lost connections.

Once the group has managed to restore the garden's senses, Deeproot will once again be able to channel information through the mycorrhizal network. The restoration of this connection symbolises the recovery of the bond between humans and the natural environment, as well as the ability to observe, understand and appreciate the biodiversity that forms part of the garden.

GAME LOOP

The game loop in Micorrizas is structured around the physical exploration of the Jardín de los Sentidos and the triggering of quests via geolocation. The experience begins on the connection screen, where one player creates the room as the host and the other participants join using a code. Once the session has been established, all players access the map scene, which serves as the central hub for navigation and group coordination.

Upon entering the map scene, the system checks the players' positions. If the group is outside the Jardín de los Sentidos, Deeproot appears to indicate that assistance can only begin once everyone has arrived at the garden. This initial condition links the start of the game to physical presence in the space and reinforces the geolocation-based nature of the experience.

Once all players are within the playable area, Deeproot introduces the main objective: to explore the garden and find the missions needed to repair the mycorrhizal network. From this point onwards, the group can move freely around the map. Progression follows a non-linear structure, as the five missions associated with the senses can be completed in any order. This decision allows the journey to adapt to the group's actual movement through the garden and avoids imposing a rigid sequence on the physical space.

The main loop is repeated in each sensory zone. Players explore the environment, enter a mission's activation area, receive the corresponding narrative context, solve the mission cooperatively, and receive feedback on their result. Upon completing the mission, the system records the group's score, marks that zone as completed, and returns the group to the map to continue their exploration. The group of adventurers can only complete each mission once. However, they can replay the entire game to achieve a higher score, as if they score less than 5, they will not have succeeded in restoring the mycorrhizal network.

Once all five missions have been completed, the game proceeds to the final screen. This scene displays the group's name, the participants' names and the group score achieved during the session. Deeproot concludes the experience by thanking the players for their help and explaining that the mycorrhizal network has restored its connections. This conclusion serves a dual purpose: it summarises the team's results and provides a narrative conclusion to the journey through the garden. These results can then be used by the course lecturers to mark the activity on which the project is based.

SCENES FROM THE GAME

The Micorrizas experience is organised into several scenes that structure the player's journey from the creation of the session to the end of the game. Each scene fulfils a specific function within the overall flow and helps to distinguish the different phases of the game: connecting the group, exploring the garden, completing missions, and presenting the final result.

The first scene corresponds to the lobby system. Here, one player creates the room as host and the other participants join using a code. This scene allows the group to form before the experience begins and ensures that all players enter the same cooperative session.

Once connected, players access the map scene. This scene represents the Jardín de los Sentidos through the integration of ArcGIS and shows the players' positions within the physical space. The map serves as the navigation hub of the experience, as it is from here that the sensory zones are detected and the corresponding missions are activated. It also allows the group to check which zones they have completed and which remain to be explored.

The mission scenes constitute the main playable part. They all share a common structure: a header at the top displays the mission objective, the relevant score or progress, and the necessary instructions so that the group knows what to do at all times. Missions are carried out cooperatively and the result obtained belongs to all members in the same room. At the end of each mission, the group receives a score between 0 and 10, which is added to the game's overall progress.

Each sensory garden has its own mission. In the Garden of Sight, players practise visual recognition of plant species using plant cards featuring an image and scientific name. In the Garden of Hearing, the group listens to sounds related to the environment and must identify them with the help of a distorted black-and-white visual clue. In the Garden of Touch, players deduce a plant from progressive clues related to physical and sensory characteristics. In the Garden of Taste, the mission uses a memory game played in pairs, themed around plants and fruits from the garden. In the Garden of Smell, players classify plants using a drag-and-drop interaction, linking them to areas such as decorative, culinary or medicinal uses.

The final scene corresponds to the final result. This screen appears when the group has completed the five main missions. It displays the group's name, the names of the participants and the group score achieved. Deeproot reappears to narratively conclude the experience and thank the players for their help in restoring the mycorrhizal network.

MECHANICS

The mechanics of Micorrizas are organised into two levels: a core mechanic common to the entire experience, and a set of specific mechanics associated with each mission. The core mechanic is the physical exploration of the Jardín de los Sentidos. Players advance by moving through the real space, and the system uses geolocation to detect when the group enters a sensory zone. Upon reaching one of these areas, the corresponding mission is triggered and the player's action shifts from the map to the challenge scene.

This main mechanic allows movement through the garden to form part of the gameplay itself. Players use the map to orient themselves, locate pending zones and check the group's progress. Geolocation links the participants' real-world position to the narrative and gameplay progression, so that each mission is tied to a specific location within the environment. All missions share several cross-cutting mechanics. The first is cooperation, as challenges are solved as a group and the score belongs to all players in the same room. The second is immediate feedback, which informs the group of the result obtained upon completing each mission. The third is shared progression, which records the areas completed and allows the group to move towards the conclusion of the experience. These mechanics maintain the game's coherence even though each mission uses a different form of interaction.

Garden of Sight In the Garden of Sight, the mission uses a card-swiping mechanic. In each round, a card appears showing an image of a plant and its scientific name. Players must decide whether that species has been observed in the Jardín de la Vista. To answer, they swipe the card to the right if they believe it belongs in that area, and to the left if they believe it does not. This interaction is well suited to mobile devices because it is quick, straightforward and easy to understand. The mission develops visual discrimination, as players must compare the information on the card with what they remember having observed in the real environment.

Garden of Hearing In the Garden of Hearing, the mission combines sound identification and cross-device cooperation. One player plays an audio clip related to the garden and the other devices display possible answers. The correct answer appears on only one of the mobiles, so the group must communicate to locate and select it. The mechanics draw on exercises involving the association of sound and word, alongside a distribution of answers among participants to reinforce collaboration. After each round, the device responsible for the audio changes, allowing different players to take on different roles during the mission.

Garden of Touch In the Garden of Touch, the mission is based on a progressive deduction mechanic. Players must write the name of a plant that might be found in that area of the garden. Each attempt generates a visual response showing partial information about the proposed species. Initially, the system reveals general characteristics, such as roughness or shape. As new attempts are made, more specific clues are unlocked, such as characteristics of the fruit, type of leaf or type of plant. This mechanic encourages comparison, the elimination of possibilities and discussion among players.

Garden of Taste In the Garden of Taste, the mission uses a memory-matching mechanic. Each device displays several face-down cards relating to fruits or plants in the garden. Players can reveal cards on their own devices, but pairs are never complete on a single mobile. To solve the mission, the group must communicate which cards each participant has discovered and coordinate to find the correct matches. This structure transforms a classic memory game into a cooperative activity distributed across devices.

Garden of Smell In the Garden of Smell, the mission uses a drag-and-drop mechanic. Players receive plant cards and must sort them into categories related to their use or function, such as decorative, culinary or medicinal. The interaction involves dragging each card to the corresponding section. This mechanic allows players to practise associating plant species with human uses, whilst maintaining an interaction that is clear and accessible on mobile devices.

GAME ART

The visual design of Micorrizas was created to support the educational and environmental tone of the project. Since the game takes place in the Jardín de los Sentidos and focuses on biodiversity, the art direction is based on a natural and friendly aesthetic. The predominant colour is green, which evokes vegetation, nature and ecological restoration. This is combined with brown and earthy tones, especially in elements related to Deeproot, trees, roots and the mycorrhizal network.

The game combines two visual approaches. On the one hand, the map has a more realistic function, as it represents the real physical space where the players move. Its purpose is to help players understand their position in the garden and connect the digital experience with the actual environment. On the other hand, the character, interface elements and icons use a simpler and cleaner style, based on flat colours, clear shapes and readable visual elements. This contrast allows the map to preserve its connection with the real location while the interface remains accessible and easy to understand on a mobile screen.

The interface was designed to be simple and direct, as the game is intended to be played outdoors. Players need to understand the information quickly while walking through the garden and interacting with the rest of the group.

Deeproot is the most important character in the visual identity of the game. His design had to communicate his connection with nature, age and wisdom, while also making him approachable as a guide for the players. The Ent's silhouette was created using Illustrator as a vector drawing, whilst the final texturing was carried out with the help of AI to achieve a bark-like appearance, complete with moss and natural details. This process resulted in a character that fits within the game's fantasy narrative, whilst maintaining a visual connection with the garden and its vegetation.

The missions also required specific visual resources. Some assets were created or adapted for the project, while others were selected from external asset libraries such as the Unity Asset Store and Itch.io. In the Garden of Hearing mission, the images used as visual clues were processed with filters so that they would not reveal the answer too directly.

Audio resources were also part of the artistic and sensory design of the game. Since the Garden of Hearing mission is based on recognising sounds related to the natural environment, sound selection was especially important. For this purpose, specialised sound databases such as Xeno-canto were used to search for accurate recordings. The selected sounds were recordings made in Spain, so that the audio material would remain coherent with the geographical and environmental context of the project.

ARCHITECTURE

The project's architecture is organised into four main blocks: connection and cooperative session, geolocation, spatial representation in ArcGIS, and the cooperative mission layer. Each block has a specific responsibility and communicates with the others via events, synchronised state structures and scene transitions.

The architecture can be described as a combination of several layers, each with a specific responsibility within the game's overall operation.

Firstly, there is a bootstrap and network connectivity layer, responsible for launching the application and enabling players to create a co-operative session or join an existing one. This layer establishes the connection between devices before the game begins.

Above this lies the session layer, which maintains the game's shared global state. In other words, this layer ensures that all players are working with the same information: which mission is active, what stage the game is at, or which elements have been completed.

The geolocation layer retrieves the device's actual position via GPS and converts it into information useful for the game. In this way, the player's physical location can be synchronised with the other participants and utilised within the co-operative experience.

Next, the map and spatial interaction layer transforms the geographical coordinates into visible and interactive elements within the game, such as markers, trigger zones or points of interest. Thanks to this layer, the real-world space of the Jardín de los Sentidos becomes part of the playable world.

Finally, the mission layer organises the game's activities. This layer is built on a common foundation that allows the general flow of missions to be reused, their results to be managed, and the transition between their different phases to be controlled.

On a practical level, this architecture relies on tools such as Unity Relay, to facilitate connections between players; Netcode for GameObjects, to synchronise information between devices; and the ArcGIS Maps SDK, to work with maps and geographical coordinates. In addition, proprietary services and drivers have been developed to clearly separate data collection, synchronisation, visual representation and the user interface.

PROBLEMS AND SOLUTIONS

During development, the main difficulties were related to geolocation, mobile testing and the integration of several technical systems into a single gameplay flow.

The first major challenge was the implementation and testing of geolocation. As the game relies on the player's position in a real-world outdoor environment, the system could not be fully validated within the Unity editor. The location service had to be tested on a real Android device, which slowed down the process and made it less predictable. To address this issue, the geolocation system was developed incrementally: first by loading and centring the map, then by limiting the rendered area, and finally by adding trigger zones for each sensory garden. The trigger zones were designed with generous radii to compensate for GPS inaccuracies.

The multiplayer system also required several design and technical decisions. The project required cross-device cooperation, but the gameplay did not require real-time character movement or complex online interaction. For this reason, the multiplayer implementation was limited to session creation, access codes, player connection, synchronisation of shared states, and group scoring. This reduced the system's complexity whilst preserving the game's cooperative structure.

A final challenge was integrating the mini-games with the rest of the application. Each mission had its own interaction model, but they all had to share the same session structure, scoring logic and progression system, so that it could be reused. To resolve this, the mini-games were connected to a common flow: activation from the map, execution of the challenge, calculation of the group score and return to the main progression.

DEVIATIONS FROM THE INITIAL PROPOSAL

The initial proposal was based on creating an escape room or gymkhana inspired by the activity suggested by the Early Childhood Education teachers. This first approach aimed to transform the guided visit into a sequence of challenges distributed throughout the "Jardín de los Sentidos", using the physical space as the basis for the player's progression.

After an initial brainstorming process with the project supervisor, the escape room structure was reconsidered. The project started to move towards a mixed reality experience in which users could learn about the biodiversity of the garden through digital information layered over the physical environment. This direction was interesting from a design perspective, since it allowed the real garden to remain central while adding interactive digital content connected to its plants, sensory areas and educational value.

However, after discussing the proposal with the Early Childhood Education teachers, the mixed reality approach was identified as difficult to apply in a realistic educational context. The main limitation was the accessibility of the required technology. A mixed reality experience would depend on specific devices that are not commonly available to students or teachers, making it harder to use the project as a practical tool during the real activity.

For this reason, the project was refocused on mobile devices, whilst ensuring that the main focus of the project remained the biodiversity of the Jardín de los Sentidos. This decision made the game more accessible and better suited for use directly in the garden, given that most potential users already own a mobile phone. The mobile platform allowed the project to maintain a connection with the physical space whilst reducing the technological barriers to the experience. From that point onwards, the design focused on developing a game for Android mobiles that could combine geolocation, cooperative interaction and educational quests within the Jardín de los Sentidos.

TESTEO Y VALIDACIÓN

ón The testing and validation phase was carried out to verify that the application worked as a functional Android prototype and that the experience was understandable and satisfactory for potential users. Since Micorrizas combines mobile interaction, geolocation, multiplayer synchronisation and outdoor exploration, validation is complex, as we need to take x users (between 2 and 6) to the Jardín de los Sentidos. Furthermore, each user must have an Android device with geolocation capability.

First, the application was tested on Android devices in order to check the main gameplay flow: creating a room, joining a cooperative session, accessing the map, detecting the player's position, activating missions, completing challenges and displaying the final result. Special attention was paid to the geolocation system, as this was the most challenging feature to get up and running; furthermore, the game relies on the player's physical presence in the Jardín de los Sentidos, as this is where the teaching activity takes place. For this reason, several tests had to be carried out in the real-world environment, as the behaviour of the GPS could not be reliably validated within the Unity editor.

In addition to technical testing, the game was tested with a group of users in order to collect feedback about the clarity of the interface, the understanding of the missions and the overall flow of the experience. During these tests, a think-aloud protocol was used. Participants were encouraged to verbalise what they were thinking while interacting with the application, which made it possible to identify confusing instructions, unclear visual elements or moments where the game flow was not sufficiently intuitive.

The first round of feedback focused mainly on the structure of the missions and the behaviour of the map. As a result of this feedback, images were added to the second mission in order to support the sound-recognition task and make the available options easier to understand. The activation areas on the map were also delimited more clearly, so that players could better understand where they needed to be in order to start a mission. What's more, when you complete the mission, there's visual feedback, the colour changes to indicate that that area of the garden has now been restored.

Other suggestions from this first round were considered but not finally implemented. One of them was the addition of a separate learning scene before the second mission. This idea was rejected because it interrupted the rhythm of the experience and added an extra step before the mission itself. Another suggestion was to prevent players from removing pairs in the fourth mission and to add an additional learning phase before it. This was also not implemented in the final version, as it would have increased the complexity of the mission and moved it away from the intended short and direct interaction model.

A second round of feedback was provided by the lecturers involved in the original educational activity. This feedback was especially relevant because it helped to adjust the prototype to the pedagogical context of the subject. In the fourth quest, an introductory reference to flavours was added so that players could better understand the relationship between the activity and the Garden of Taste. In the first quest, the plant images were enlarged to make visual recognition easier on mobile screens. In the fourth mission, an additional pressure element was introduced to make the activity more dynamic. The user interface was redistributed in some scenes to improve readability and interaction, and the avatar of the GPS indicator was adjusted so that players could more easily interpret the location state of the application.

The validation process helped to refine the game from both a technical and an educational point of view. The tests confirmed that the application could run as an Android prototype and that the main cooperative flow was functional. At the same time, the feedback received made it possible to improve the clarity of the missions, the readability of the interface and the connection between the game mechanics and the educational activity on which the project is based.

RESULTS

The final result of the project is a functional Android application that transforms the educational activity of the Jardín de los Sentidos into a cooperative, geolocated serious game. The application allows players to create a multiplayer session, join it using a code and take part in a shared experience in which the group explores the real garden and completes missions linked to its sensory areas.

The application includes the main systems required to support the intended experience. It integrates geolocation so that the player's real position can be used inside the game, a map scene that acts as the central navigation space, a cooperative multiplayer structure that keeps the session synchronised between devices, and a set of missions connected to the senses represented in the garden. The game also includes feedback after each mission and a final screen showing the group name, the participants and the score achieved during the session.

From a gameplay perspective, the result is a complete playable flow. Players can start a session, access the map, move through the garden, activate missions, solve them cooperatively and obtain a final result. The experience is structured around exploration, observation, communication and group decision-making, maintaining the relationship with the physical environment throughout the game.

From an educational perspective, the application provides the lecturers with a digital tool that can be used to support the teaching session related to the Jardín de los Sentidos. The final score and the group result can be used as part of the activity carried out in class, allowing the teachers to assess participation and performance during the session. In this sense, the project has produced a usable prototype that can be incorporated into the educational context for which it was designed.

PROJECT APPLICATIONS

Micorrizas has been developed as a tool that can be applied within the subject related to the natural environment in Early Childhood Education. Its main application is to complement the existing visit to the Jardín de los Sentidos by providing a structured, interactive and cooperative activity that students can complete in the real space. If any problems arise, such as poor lighting because the activity is taking place late in the day or adverse weather conditions.

The game can be used by lecturers to guide students through the garden without requiring constant direct instruction. The application gives players objectives, activates quests according to their position and provides feedback on their answers. This allows the teaching activity to become more autonomous while preserving the importance of direct contact with the natural environment.

Another possible application is its use as an assessable class activity. Since the game records the group's progression and provides a final score, the lecturers can use the result as part of the evaluation of the session. This does not replace the educational value of observation and discussion, but it provides an additional element that can help structure and assess the activity.

The project also creates a foundation for future educational uses. New plant species, sounds, questions, clues or missions could be added to adapt the experience to future editions of the subject or to different learning objectives. Therefore, the application is not limited to the current application, but can be expanded as a reusable educational resource.

CONCLUSIONS AND FUTURE WORK

CONCLUSIONS

The development of Micorrizas has made it possible to create a functional Android serious game based on a real educational activity. The project has achieved its main technical objective by producing a mobile application that combines geolocation, cooperative multiplayer interaction, mission-based gameplay and a final group result. These elements work together to support an outdoor learning experience in the Jardín de los Sentidos.

One of the main conclusions of the project is that the game is not simply a gamified version of an existing activity. During development, it became clear that the original visit was being used as the basis for designing a serious game with its own structure, narrative, mechanics and progression. The educational activity provided the context, contents and objectives, but the final result required specific design decisions in order to transform that material into an interactive experience.

The project has also shown the value of connecting mobile gameplay with a real environment. By using the garden as the playable space, the game encourages players to observe, move, communicate and make decisions in relation to the physical surroundings. This supports the objective of promoting awareness and appreciation of biodiversity through interaction with the natural environment.

However, the educational impact of this objective cannot yet be fully measured. The application is prepared to be introduced as a tool in the subject, and the lecturers have expressed the intention of using it in a real teaching session during the next academic year. Once the game is used with students in that context, it will be possible to evaluate more precisely whether the experience increases their awareness, appreciation and understanding of the biodiversity of the Jardín de los Sentidos.

Working on mycorrhizae has enabled me to put much of the knowledge I acquired during my degree into practice, as well as to incorporate new knowledge that has proved necessary during the course of the project.

Overall, the result achieved is satisfactory, as the game meets the project's expectations and demonstrates the feasibility of building a geolocation-based cooperative experience in Unity.

FUTURE WORK

The main line of future work is the use of the application in a real session of the subject. This would allow the game to be tested with its intended users and under authentic teaching conditions. The results of that implementation could be used to evaluate the educational impact of the game, especially in relation to student engagement and collaborative learning.

A future evaluation could combine several methods, such as observation during the activity, questionnaires before and after the session, teacher feedback and analysis of the final group scores. This would make it possible to assess not only whether the application works technically, but also whether it contributes to the learning objectives of the original activity.

From a technical perspective, future work could focus on improving GPS accuracy and robustness during outdoor use. Although the prototype already includes geolocation-based mission activation, further testing with larger groups and different mobile devices would help to refine activation areas, GPS indicators and tolerance margins.

The content of the game could also be expanded. New plant species, sounds, images and questions could be added to enrich the missions and make the experience more varied. Additional missions could also be designed for other botanical or educational spaces, allowing the structure of Micorrizas to be reused beyond the Jardín de los Sentidos.

Finally, the teacher-facing part of the application could be improved. Future versions could include tools to export results, review group performance or configure the content of the missions before the session. These improvements would make the application more useful as a teaching resource and would support its integration into the subject in a more stable way.

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APPENDIX

SOURCE CODE
SIMULACIÓN EN LOS MÓVILES
ARCHITECTURE